

3. Repeated eigenvalues

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Deriving the independent solutions for repeated eigenvalues

2/2 points (graded)

Let

$$\mathbf{A} = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \quad \text{where } b \neq 0. \tag{3.115}$$

Find two independent solutions to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ by solving the two differential equations separately. That is, find $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ such that the general solution is

$$\mathbf{x}(t) = C_1 \mathbf{x}_1(t) + C_2 \mathbf{x}_2(t) \quad \text{where } C_1, C_2 \text{ arbitrary constants.}$$

Calculator

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Write down the solution $\mathbf{x}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$ to the initial value problem with initial conditions $\mathbf{x}(0) = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$. (Your answer should be in terms of x_0 , y_0 , and b .)

$x(t) =$

y_0*b*t*e^t+x_0*e^t

y_0 · b · t · e^t + x_0 · e^t

$y(t) =$

y_0*e^t

y_0 · e^t

Use the mathlet below to look at the phase portrait of the system.

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How and why to use the forum

While answers to most individual problems are provided, staff do not provide detailed solutions to recitation problems. Instead, use the forum to collaborate with your fellow students to develop detailed solutions for these problems.

The goal is to get practice developing your own detailed solutions, practice explaining solutions to others, and read other people's solutions and reflect on their process and reasoning.

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Topic: Unit 2: The geometry of solutions / 3. Repeated eigenvalues

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My attempt

discussion posted 3 months ago by Hryhorchuk

We are solving a system: $\begin{cases} \dot{x} = x + by \\ \dot{y} = y \end{cases}$ The second equation can be solved immediately: $y = C_1 e^t$. Plugging this solution into the first equation: $\dot{x} = x + C_1 b e^t$. This is a non-homogeneous linear DE with exponential input, and we can get a particular solution either by variation of parameters or by ERF/ERF: $x_p = C_1 b t e^t$. We can solve $\dot{x} = x$ to get the homogeneous solution: $x_h = C_2 e^t$. So the general solution is $x = C_1 b t e^t + C_2 e^t$. Rewriting solutions to both equations in matrix form:
$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} C_1 b \\ 0 \end{pmatrix} t e^t + \begin{pmatrix} C_2 \\ C_1 \end{pmatrix} e^t = C_1 \begin{pmatrix} b t \\ 0 \end{pmatrix} e^t + C_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^t + C_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t = C_1 \begin{pmatrix} b t \\ 1 \end{pmatrix} e^t + C_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t$$
 Plugging the initial conditions: $\begin{pmatrix} b \cdot 0 & \text{amp;} 1 \\ 1 & \text{amp;} 0 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$, which is solved by $C_1 = y_0, C_2 = x_0$ Then the solution becomes: $\begin{pmatrix} x \\ y \end{pmatrix} = y_0 \begin{pmatrix} b t \\ 1 \end{pmatrix} e^t + x_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t$, which is the same as $x(t) = x_0 e^t + y_0 b t e^t, y(t) = y_0 e^t$.

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2 responses

jfrench (Staff)

3 months ago

I just made a little edit. Looks like \ needs to be escaped as \\! :)

Thanks Jent!

I am really confused by the new forum features. Already created 2 reports to edX support about the bugs.

posted 3 months ago by Hryhorchuk

I am also really confused by the new forum, and why it is behaving differently in different places! So confusing!

posted 3 months ago by jfrench (Staff)

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MarkvdV

3 months ago

That's a cool solution.

I plugged in $y(t) = y_0 e^t$ into the differential equation for x , which resulted in the general solution $x = C_1 e^t + C_2 b y_0 t e^t$. I determined C_1 and C_2 by using the initial conditions $x(0) = x_0$ and $\dot{x}(0) = x_0 + b y_0$.

Cheers

Is the vector space of solutions 1-dimensional or 2-dimensional? In "9. On the critical parabola" we see that in the case of a degenerate node we have a 1-dimensional space of eigenvectors, but in the solution here we have got two different eigenvectors.

posted 3 months ago by Marco_Str

We are dealing with a 1-dimensional eigenvector space in this example. Matrix $A = \begin{pmatrix} 1 & \text{amp;} b \\ 0 & \text{amp;} 1 \end{pmatrix}$, so $\lambda^2 - 2\lambda + 1 = 0$. Therefore, $\lambda = 1$, repeated.

Now, to determine the eigenvector: $\begin{pmatrix} 0 & \text{amp;} b \\ 0 & \text{amp;} 0 \end{pmatrix} \vec{v} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The eigenvector $\vec{v} = \begin{pmatrix} a \\ 0 \end{pmatrix}$, where a can be any number.

This means that the basis of the vector space consists of 1 vector, for example $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, which makes it a 1-dimensional vector space.

posted 3 months ago by MarkvdV

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